

The advantage is complexity reduction and rendering speed, because only one texel read is required for each pixel being written to the frame buffer. The disadvantage is the blocky image that results when the object moves.

Transformation

A change of coordinates; a series of mathematical operations that act on output primitives and geometric attributes to convert them from modelling coordinates to device coordinates.

Tri-linear MIP Mapping

A method of reducing aliasing artefacts within texture maps by applying a bilinear filter to four texels from the two nearest MIP maps and then interpolating between the two.

Z-buffer

A part of off-screen memory that holds the distance from the view-point for each pixel—the Z-value. When objects are rendered into a 2D frame buffer, the rendering engine must remove hidden surfaces.

Z-buffering

A process of removing hidden surfaces using the depth value stored in the Z-buffer. Before bringing in a new frame, the rendering engine clears the buffer, setting all Z-values to infinity. When rendering objects, the engine assigns a Z-value to each pixel: the closer the pixel to the viewer, the smaller the Z value. When a new pixel is rendered, its depth is compared with the stored depth in the Z-buffer. The new pixel is written into the frame buffer only if its depth value is less than the stored one.